

scripts/run-vm-distro-matrix.sh — User Guide

Last verified: 2026-05-16 (Phase 6a + 6b §11.4.29 snake_case migration: scripts/run-vm-distro-matrix.sh path-references updated from Submodules/ to submodules/) **Inheritance:** HelixConstitution §11.4.18 (script documentation mandate)

Overview

Stub doc generated from the script's in-source header comment. See scripts/run-vm-distro-matrix.sh for canonical behavior. This stub exists so the CM-SCRIPT-DOCS-SYNC pre-push gate (.github/hooks/pre-push Check 9) starts gating modifications to this script.

The script's own header documentation (verbatim):

```
scripts/run-vm-distro-matrix.sh - cross-OS distro matrix wrapper.
Drives cmd/vm-matrix against (alpine, debian, fedora) × x86_64 = 3
configs. Each VM starts proxy.jar + lava-api-go in background and
probes 4 endpoints (proxy_health, proxy_search, goapi_health,
goapi_metrics). Post-processing rejects when any of 4×3=12
booleans
is false.
```

Bluff vector this catches: distro-specific runtime behavior – a proxy or Go binary that runs cleanly on alpine but fails on debian (musl vs glibc, systemd vs OpenRC, packaged JRE differences). The Sixth Law clause 4 acceptance gate ("works for a real user end-to-end") fails silently if we only test on one distro family.

Pre-requisites:

- proxy/build/libs/app.jar exists (./gradlew :proxy:buildFatJar)
- lava-api-go/build/lava-api-go binary exists
- tools/lava-containers/vm-images.json has REAL hashes

(placeholder

zeros reject in pkg/cache.Store.Get())

Exit codes:

- 0 – all 4 probes succeeded on all 3 distros (12/12 true)
- 1 – at least one probe failed OR a probe-output.json is missing
- 2 – configuration error (missing inputs)

Usage

See the script's in-source comment block (above) for canonical usage examples.

Maintenance

When this script is modified, update this document in the same commit (CM-SCRIPT-DOCS-SYNC requires it). Per §11.4.18, the documentation **MUST** stay in sync with the codebase — no doc may be out of sync with its script.

Cross-references

- `scripts/run-vm-distro-matrix.sh` — the script itself
- `docs/helix-constitution-gates.md` — gate inventory
- `HelixConstitution Constitution.md` §11.4.18 (the mandate)
- `Lava CLAUDE.md` §6.AD (HelixConstitution Inheritance)